Predicting the efficacy of malaria vaccines across the life-course

Supplementary Notebook 10. Sensitivity analysis

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Here we demonstrate the sensitivity of the model outputs of lifetime deaths averted and lifetime death efficacy against blood infection rate to changes in the model parameters. The two outputs we focus on are:

1. the critical infection rate, λ_RH5, at which the combination of R21+RH5 transitions from under-performing to out-performing RH5-alone (R21+RH5 always out-performs R21-alone)
2. the critical infection rate, λ_R21, at which the R21-alone vaccine with one booster transitions from indirectly causing deaths to preventing deaths.

For the default parameters of the model we find $\lambda_\text{R21} \approx 1.5$ infections per year and $\lambda_\text{RH5} \approx 3.5$ infections per year.

The sensitivity analysis below shows that

- parameters that substantially affect λ_RH5 are:

rho_elp epsi_severe omega_clinical

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- Decay rate age-modifier in risk of severe malaria, $\epsilon_\text{severe}$
- RH5 protective effect against clinical malaria, $\omega_\text{clinical}$
- RH5 half-life, $\tau_\text{blood}$
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- parameters that substantially affect λ_R21 are:
  - Decay rate age-modifier in risk of severe malaria, ϵ_severe
  - R21 half-life, τ_liver

Although both critical infection rates λ_R21 and λ_RH5 depend sensitively on ϵ_severe, its value is quite constrained around 0.05 per year per infection. If it is much lower than 0.05 a second peak in deaths in 5 to 10 year olds occurs at low blood infection rates, which is not observed in the epidemiological data. If it is much higher than 0.05 it overly suppresses deaths in the 5 to 10 year olds, which do occur.

RH5's protective effect against clinical malaria, ω_clinical, needs to be determined as it is currently unknown. This will probably only be achieved by counting all infections (clinical and asymptomatic) in the control and vaccine arms of future trials.

1 Early life protection parameters

Early life protection from infection wanes from a maximum ρ_elp at birth to zero with a half-life of τ_elp years.

1.1 Early life protection from infection at birth, ρ_elp

Early life protection from infection at birth, ρ_elp, is estimated to be 0.71 in the Senegalese villages of Ndiop and Dielmo (Notebook S5; Trape and Rogier (1996); Trape et al. (2024)). In Figure 1 ρ_elp varies from 0.5 to 1 (default is 0.71).

Changing this parameter is similar to changing infection rate by a small amount, which causes the curves to shift horizontally. It has a small quantitative effect on λ_{R21} and λ_{RH5}. But qualitatively it has no effect on the conclusions of the paper.

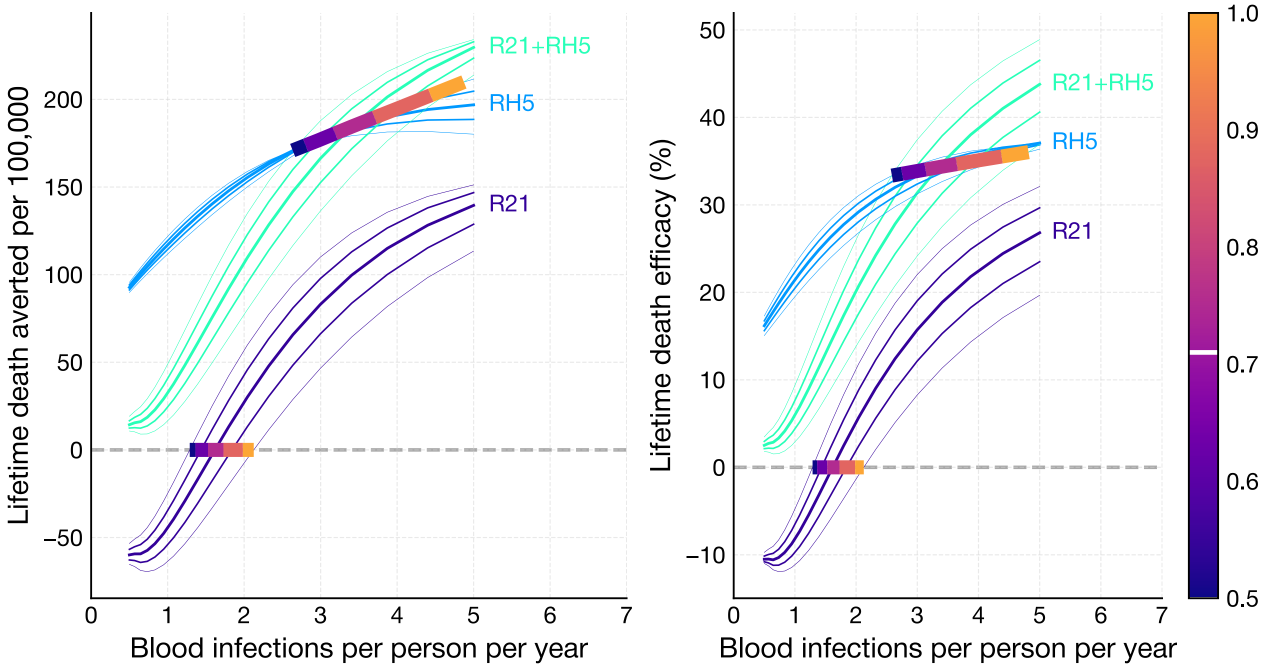

Figure 1: Early life protection from infection at birth, $\rho_{\mathrm{elp}} = (0.5, 1)$. The default value is shown as a white line in the colorbar. For a given value of ρ_{elp} and for each vaccine strategy (R21-alone, RH5-alone and RH5+R21 combined, all with one booster) we run 20 simulations across the range of infection rates from 0.5 to 5 blood infections per year. The point of intersection of the R21+RH5 line and the RH5 line is the critical threshold λ_{RH5}. And the point of intersection of the R21 line with the x-axis is the critical threshold λ_{R21}. We repeat the process for five different values of ρ_{elp} equally spaced between 0.5 and 1 inclusive. The resulting five values of λ_{RH5} are joined by a coloured line with the colour representing the value of ρ_{elp} which is shown in the colorbar. The same is done for λ_{R21}. Thus the coloured lines show the sensitivity of the two critical thresholds to changes in ρ_{elp}.

1.2 Half-life of early life protection from infection, τ_{elp}

Half-life of early life protection from infection, τ_{elp}, is estimated to be 1.70 years in the Senegalese villages of Ndiop and Dielmo (Notebook S5, Trape and Rogier (1996), Trape et al. (2024)). The infection data in Figure 2A of Gonçalves et al. (2014), suggests that early life protection wanes within the six months of life. In Figure 2 it varies from 0.5 to 2.5 years (default is 1.70 years).

As for ρ_{elp}, changing this parameter is similar to changing infection rate. Again the quantitative effects on λ_{R21} and λ_{RH5} are small but with no qualitative effect.

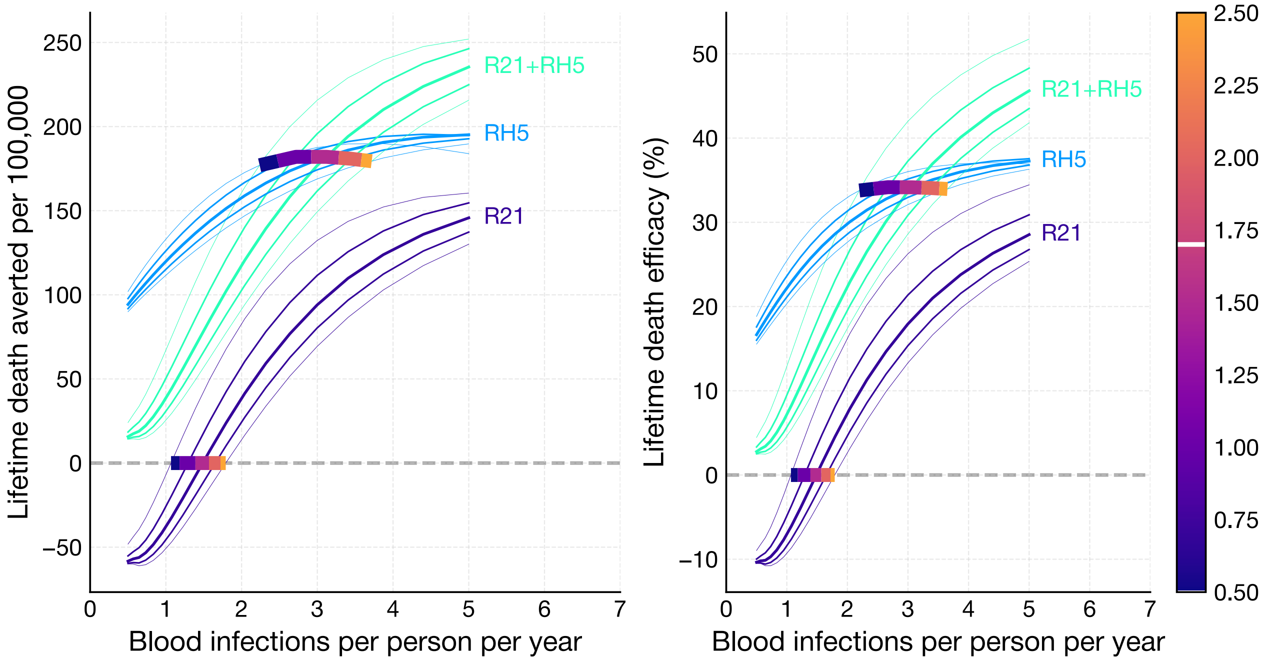

Figure 2: Half-life of early life protection from infection, $\tau_{\text{elp}} = (0.5, 2.5)$ years.

2 Clinical immunity parameters

The risk of clinical malaria declines sigmoidally with exposure (i.e., cumulative number of infections). On first infection the relative risk is given by ρ_{clinical}. The half-maximum-risk exposure is given by γ_{clinical}, and the rate of decline of risk at half-maximum-risk is proportional to δ_{clinical}.

2.1 Relative risk of clinical malaria on first infection, ρ_{clinical}

Relative risk of clinical malaria on first infection, ρ_{clinical}, is estimated to be 0.8 (Notebook S4, Gonçalves et al. (2014)). In the individual-level data for Dielmo village it varied from about 0.2 to 1 (Notebook S5, Trape and Rogier (1996), Trape et al. (2024)). In Figure 3 it varies from 0.20 to 1 (default is 0.79).

This parameter scales clinical malaria age patterns but does not alter infection rate. It therefore affects the number of averted deaths (fewer clinical episodes implies fewer severe malaria episodes and deaths) but it does not affect the critical infection rates λ_{R21} and λ_{RH5}. Moreover, because it affects unvaccinated and vaccinated children equally, death efficacies are unchanged.

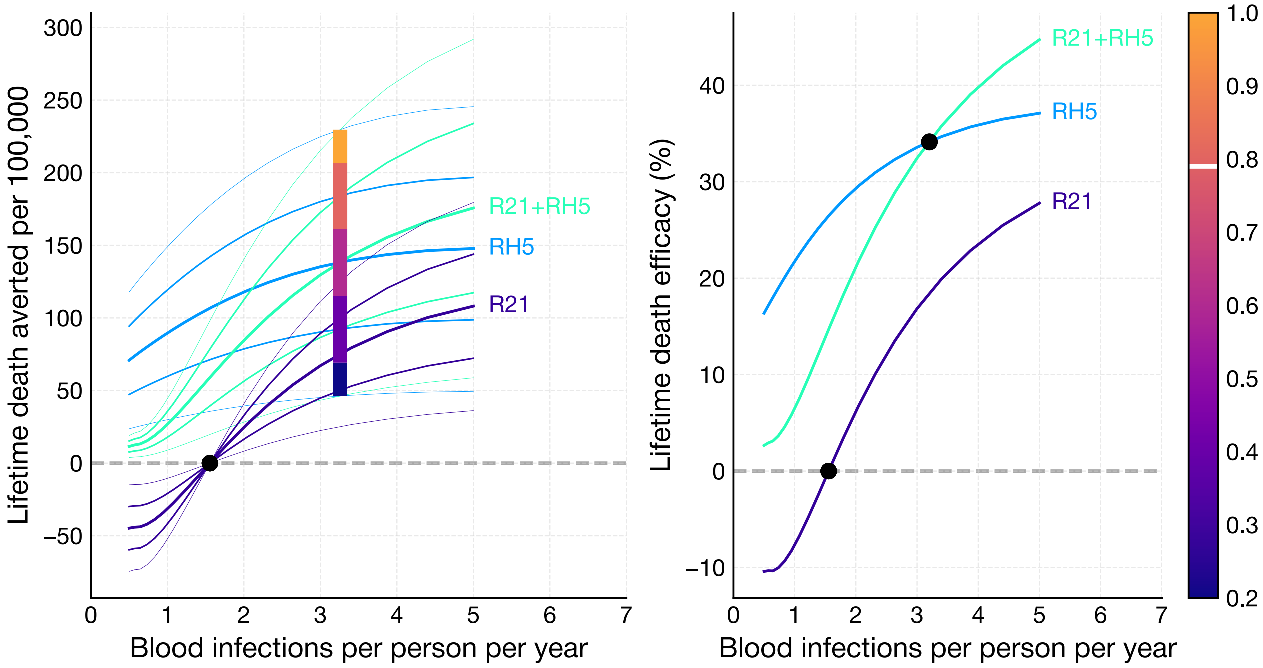

Figure 3: Relative risk of clinical malaria on first infection, $\rho_{\text{clinical}} = (0.20, 1)$.

2.2 Half-maximum-risk exposure of clinical malaria, γ_{clinical}

Half-maximum-risk exposure of clinical malaria, γ_{clinical} is estimated to be 57.00 infections for the population-level data of Dielmo and Ndiop villages. In the individual-level data for Dielmo village it varied from about 10 to 200 infections (Notebook S5, Trape and Rogier (1996), Trape et al. (2024)). In Figure 4 it varies from 10 to 200 infections (default is 57.00 infections).

This parameter changes clinical age patterns in older children so it has negligible effect on severe and death age patterns and negligible effect on the critical infection rates.

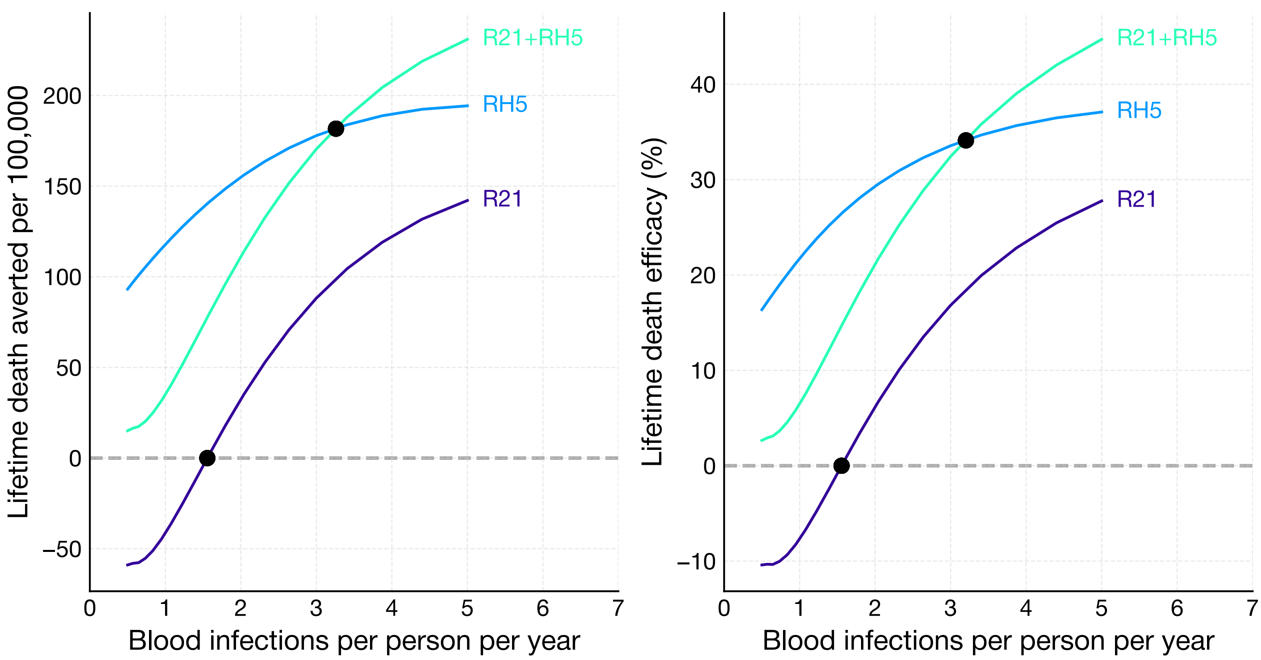

Figure 4: Half-maximum risk exposure of clinical malaria, $\gamma_{\text{clinical}} = (10, 200)$ infections.

2.3 Rate of development of clinical immunity with exposure, δ_{clinical}

Rate of development of clinical immunity with exposure, δ_{clinical}, is estimated to be 0.03 per infection for Dielmo and Ndiop villages. In the individual-level data for Dielmo village it varied from about 0.01 to 0.06 per infection (Notebook S5, Trape and Rogier (1996), Trape et al. (2024)). In Figure 5 it varies from 0.01 to 0.1 (default is 0.03 per infection).

This parameter changes clinical age patterns in older children so it has negligible effect on severe and death age patterns and negligible effect on the critical infection rates.

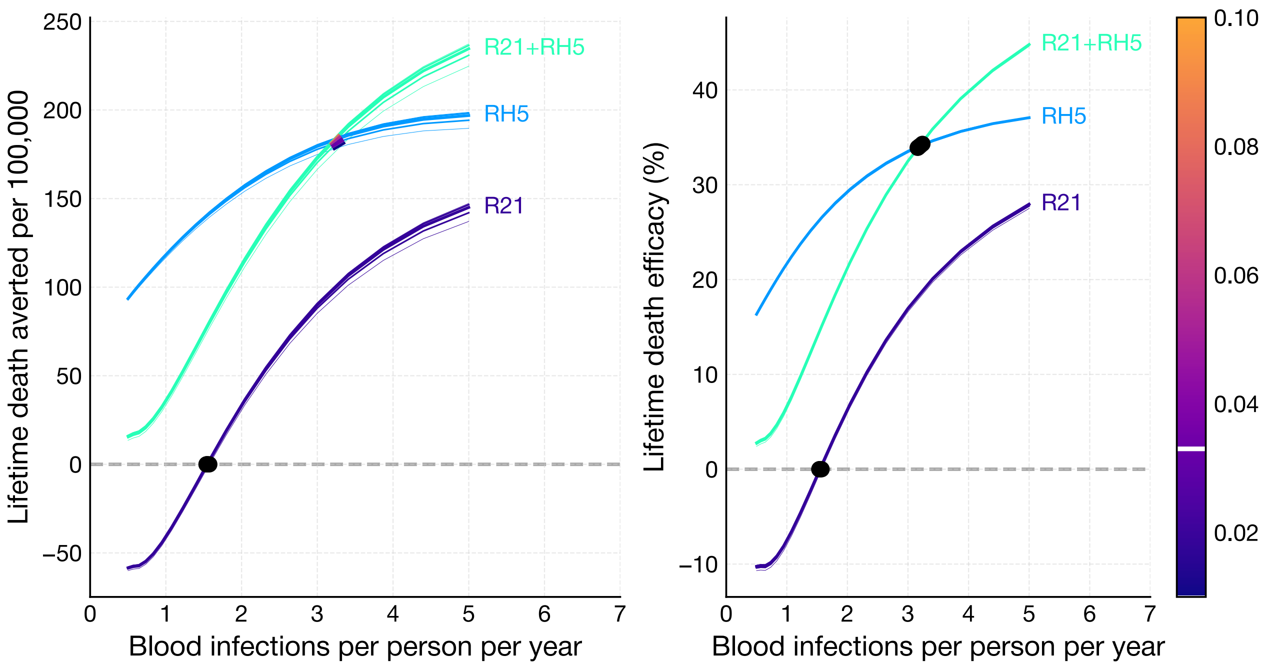

Figure 5: Rate of development of clinical immunity with exposure, $\delta_{\text{clinical}} = (0.01, 0.1)$ per infection.

3 Severe malaria immunity parameters

The risk of a severe episode becoming severe declines exponentially with age-dependent exposure. On first infection, the relative risk is ρ_{severe}. The rate of decline in risk is $\delta_{\text{severe}} + \epsilon_{\text{severe}} a$ where δ_{severe} is the rate in decline of risk in newborns and ϵ_{severe} is an age-modification parameter that increases the rate of decline in risk with age a.

3.1 Relative risk of severe malaria on first infection, ρ_{severe}

Relative risk of severe malaria on first infection, ρ_{severe} is estimated to be 0.04 (Notebook S4; Gonçalves et al. (2014); Muñoz Sandoval et al. (2025)). In Figure 6 it varies from 0.01 to 0.08 (default 0.04).

This parameter scales the number of severe malaria episodes without changing the shape of severe malaria and death age patterns. So deaths averted is scaled, but death efficacy is left unchanged.

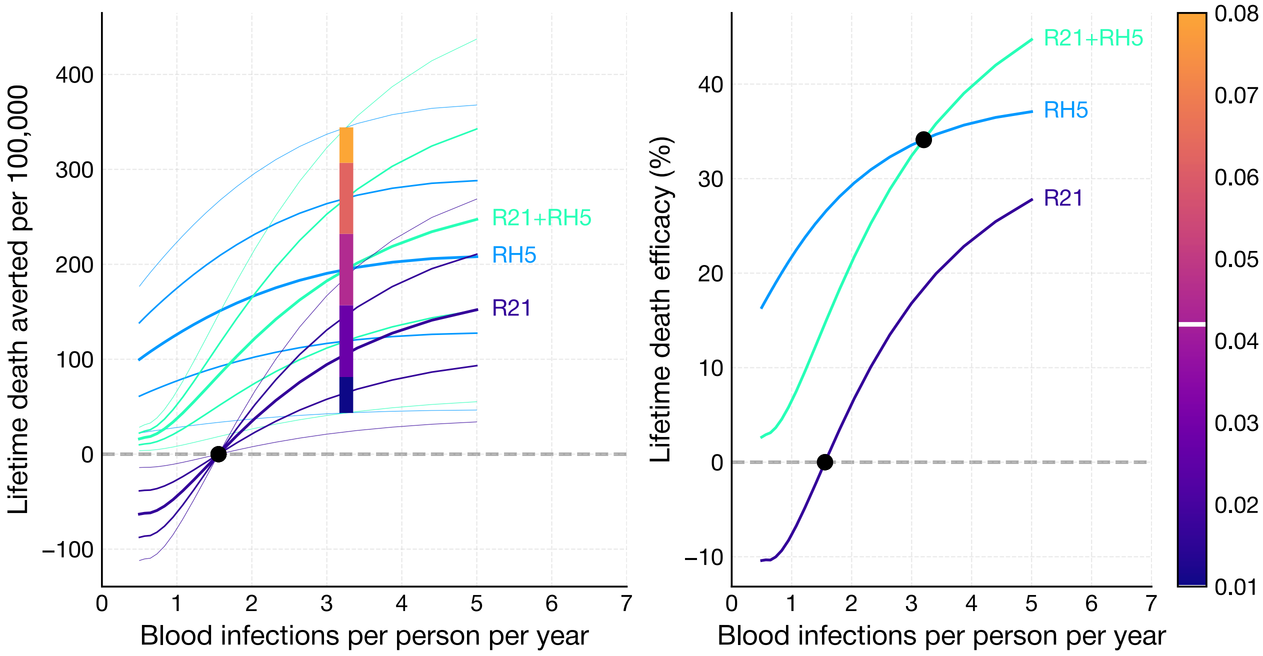

Figure 6: Relative risk of severe malaria on first infection, $\rho_{\text{severe}} = (0.01, 0.08)$.

3.2 Rate of decline in risk of severe malaria with exposure in newborns, δ_{severe}

Rate of decline in risk of severe malaria with exposure in newborns, δ_{severe} is estimated to be 0.12 per infection (Notebook S4, Gonçalves et al. (2014), Muñoz Sandoval et al. (2025)). In Figure 7 it varies from 0.05 to 0.25 per infection (default 0.12 per infection).

This parameter changes the rate of development of severe immunity. Hence increasing this parameter causes the severe malaria age pattern to peak at earlier ages thereby reducing overall CFR which in turn reduces the impact vaccines have on averted deaths. The effect on λ_{R21} is negligible. The effect on λ_{RH5} is small.

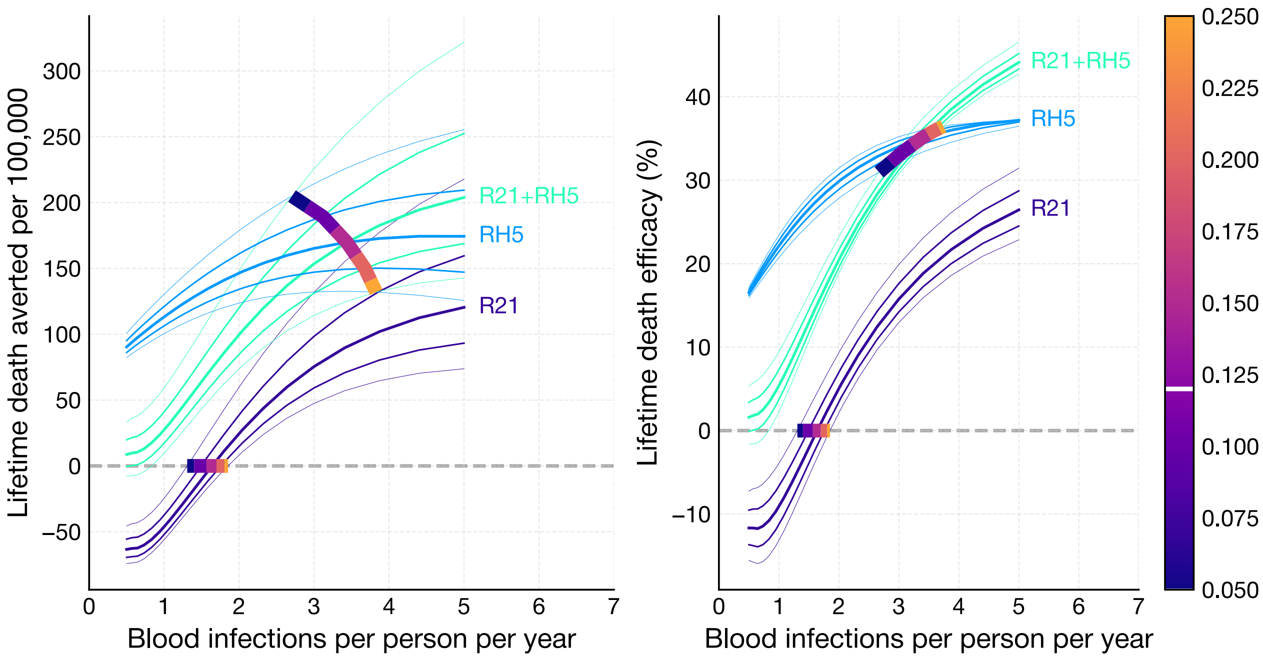

Figure 7: Rate of decline in risk of severe malaria with exposure in newborns, $\delta_{\text{severe}} = (0.05, 0.25)$ per infection.

3.3 Decay rate age-modifier in risk of severe malaria, ϵ_{severe}

Decay rate age-modifier in risk of severe malaria, ϵ_{severe} is estimated to be about 0.05 per year per infection (Notebook S6). This estimate is not based on formal fitting to data, but on its ability to recapitulate mortality age patterns (Abdullah et al. 2007). There is, therefore, some uncertainty in its value. However, values lower than 0.05 cause a second peak in deaths in 5 to 10 year olds, which is not observed in epidemiological data. Values higher than 0.05 tend to suppress deaths in the 5 to 10 year olds, which are known to occur (see for example, Abdullah et al. (2007), Reyburn et al. (2005)). In Figure 8 it varies from 0.035 to 0.75 per year per infection (default 0.05 per year per infection).

The effect of this parameter on the averted deaths and efficacies curves is complex. And λ_{R21} and λ_{RH5} are both sensitive to its value.

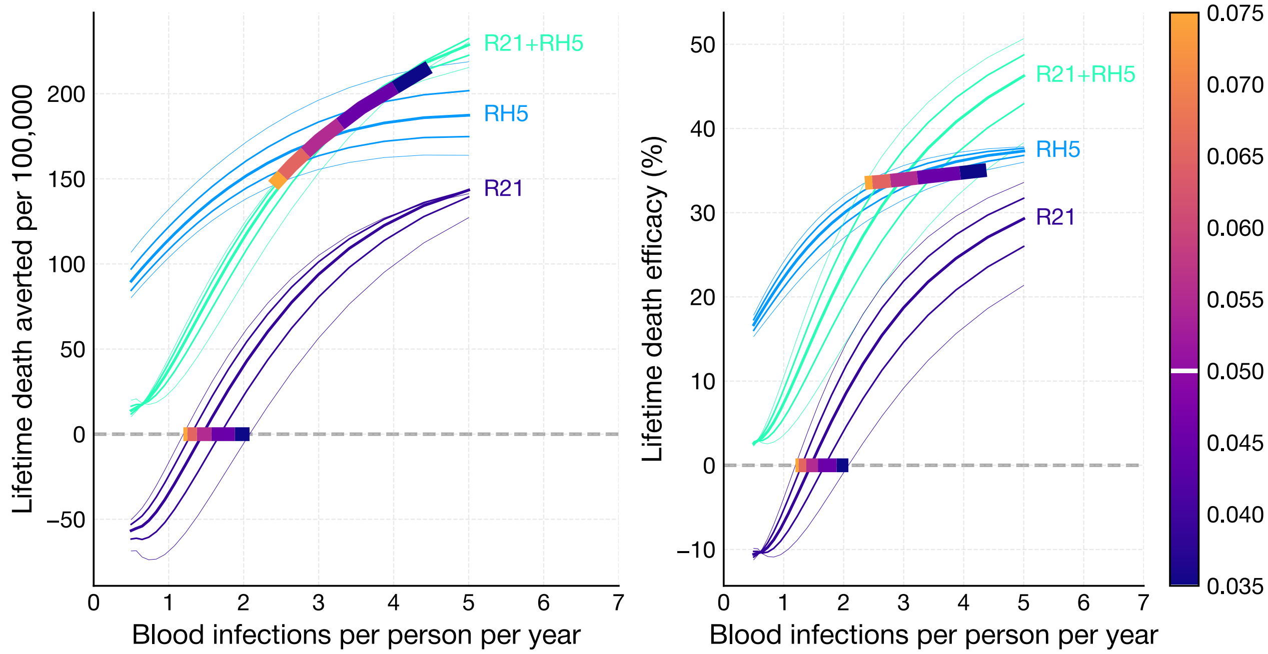

Figure 8: Age modification rate in decline in risk of severe malaria, ϵ severe $= (0.035, 0.075)$ per infection per year.

4 Strength of severe malaria-associated age-specific risk of death

Severe malaria-associated age-specific risk of death is estimated from Reyburn et al. (2005). The default risk is shown as a dashed line in Figure 9 (see Notebook S7). The model parameter `cfr_modifier` attenuates the strength of age-specificity. A value of 1 corresponds to the blue dashed line in Figure 9. A value of zero completely removes age-specificity and the risk of death is a constant 0.067 across all ages (green line). This is the mean risk of death in the Reyburn et al. (2005) data.

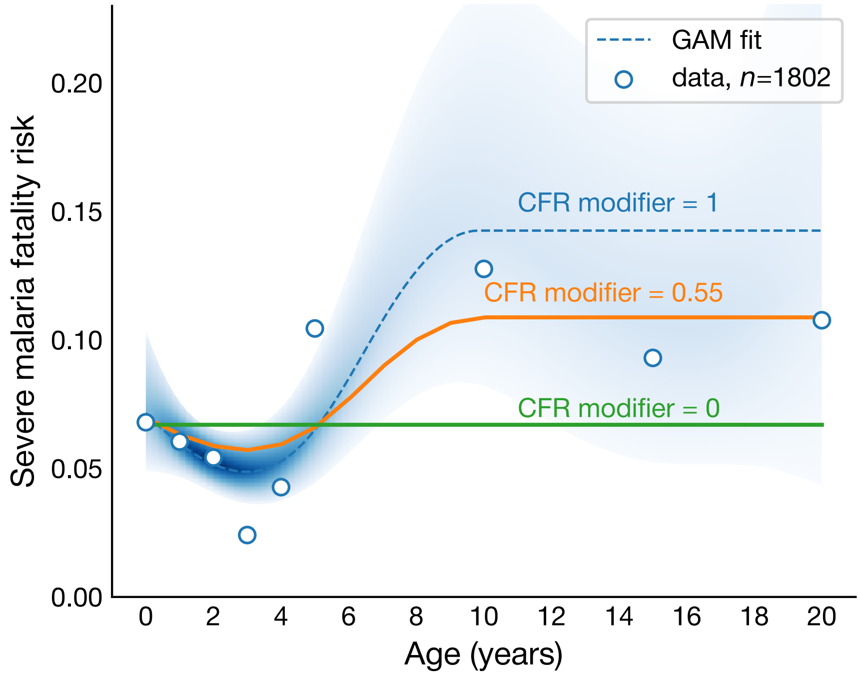

Figure 9: Estimated severe malaria-associated mean age-specific risk of death (dashed line) with 95% CI (blue band) and age-stratified data (circles) from Reyburn et al. (2005).

In Figure 10 `cfr_modifier` varies from 0.55 to 1 (default is 1.0).

Attenuating the age-specificity of the risk of death means that children aged over five have more similar CFR to children under five. So the rebound in deaths in the over fives caused by R21 is weakened. This causes the R21 curves in Figure 10 to move upward resulting in λ_{R21} moving toward zero. For `cfr_modifier` less than about 0.6 λ_{R21} is zero and therefore R21 cannot indirectly cause deaths.

Similarly for the combined R21+RH5.1 vaccine, rebound is weakened causing the R21+RH5.1 curves to move upward and λ_{RH5} to decrease.

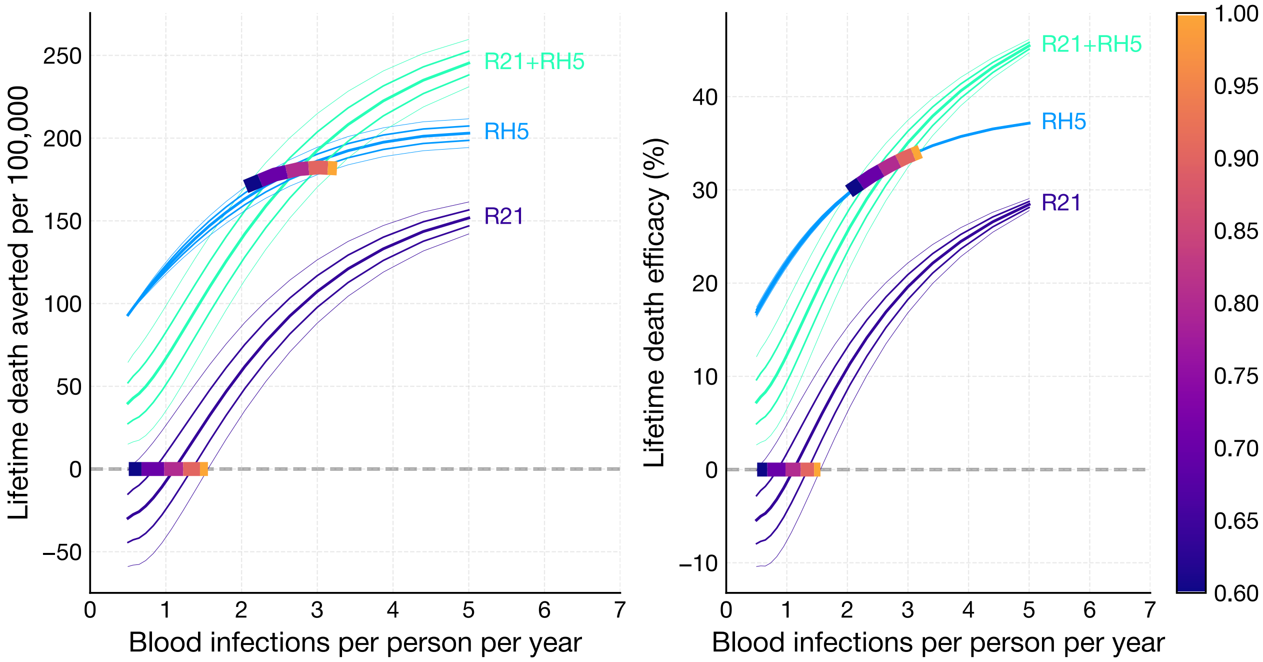

Figure 10: Attenuation of severe malaria-associated age-specific risk of death, `cfr_modifier` = (0.6, 1).

5 Blood infection rate heterogeneity, β

The parameter β determines the degree of heterogeneity in blood infection rate in the population, with 0 being completely homogeneous, and 1 being maximally heterogeneous (modelled assuming a symmetric triangular distribution of infection rates from 0 to twice the mean annual infection rate, see Rodriguez-Barraquer et al. (2016)). In Figure 11 it varies from 0 to 1 (default is 1).

Its effect on lifetime averted deaths and efficacy is negligible and, therefore, its effect on the two critical infection rates is also negligible.

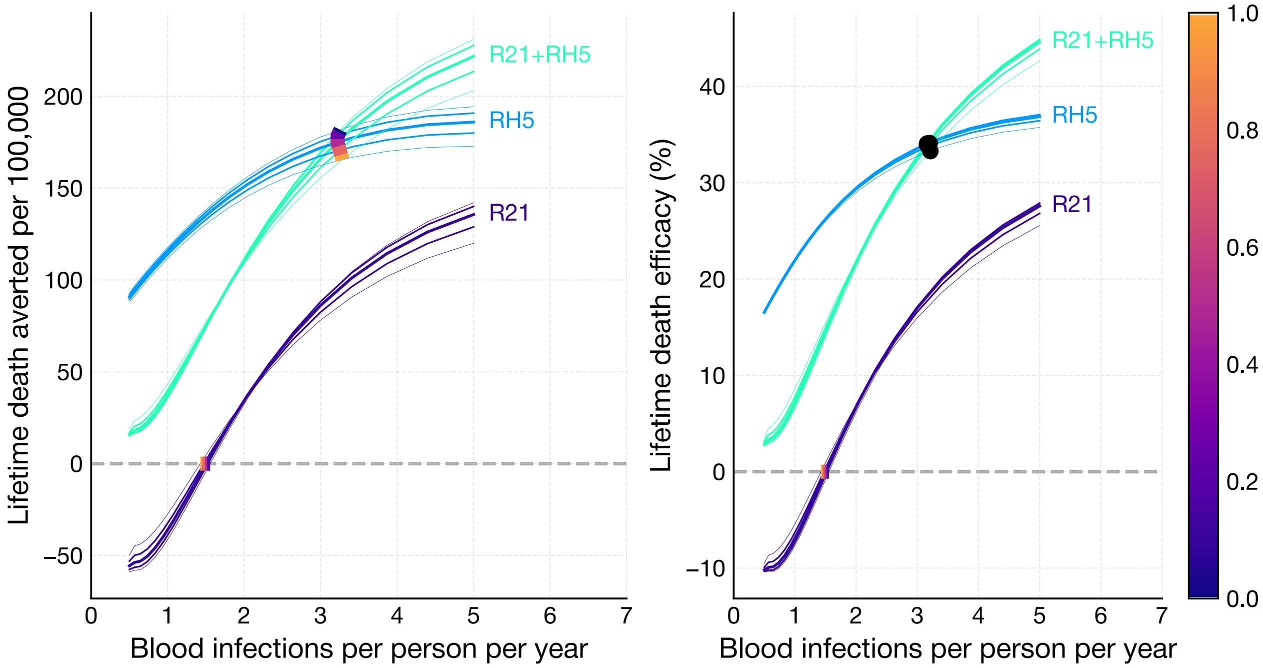

Figure 11: Blood infection rate heterogeneity, $\beta = (0, 1)$.

6 R21 vaccine parameters

The R21 protection profile against blood infection we use is estimated in Schmit et al. (2024). There are two parameters we can change that affect this profile: the maximum efficacy just after the last primary dose ν_{liver}, and the vaccine's half-life τ_{liver}. We can also change the number of yearly boosters.

6.1 R21's maximum efficacy against infection, ν_{liver}

R21's maximum efficacy, ν_{liver} scales protective efficacy against infection, where $\nu_{\text{liver}} = 1$ corresponds to the same efficacy profile as in Schmit et al. (2024). In Figure 12 it varies from 0.5 to 1 (default is 1.00).

R21 temporarily reduces infection rate which delays infection, but it does not directly prevent death. Whether it indirectly prevents or causes deaths depends on how late into childhood infection is delayed due to the opposing forces of the development of age-dependent severe malaria immunity and age-dependent CFR. The parameter ν_{liver} can have a negative or positive impact on deaths averted and death efficacy depending on infection rate (Figure 12). (Obviously, this parameter has no effect on the RH5-alone vaccine curves.) Whatever its value though, its impact on the critical infection rates is minimal.

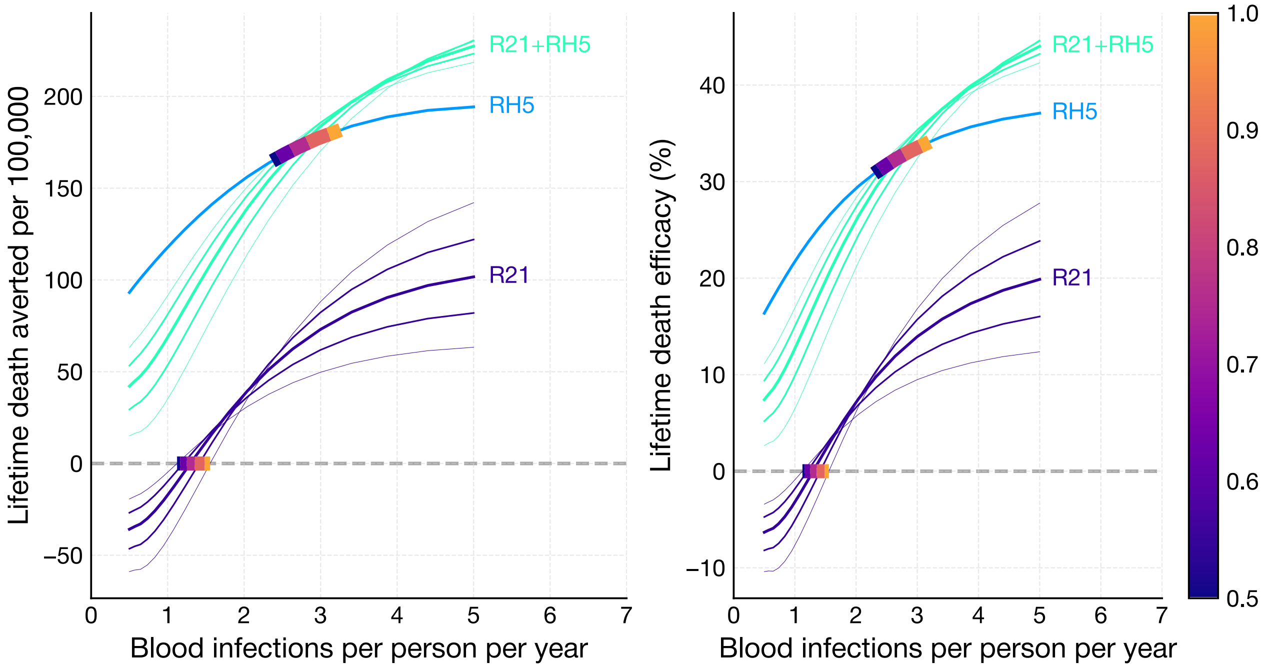

Figure 12: R21's maximum efficacy against infection, $\nu_{\text{liver}} = (0.5, 1)$.

6.2 R21 half-life, τ_{liver}

The parameter τ_{liver} scales the half-life of R21, where $\tau_{\text{liver}} = 1$ corresponds to the same efficacy profile as in Schmit et al. (2024). R21 is estimated to have a long half-life of over five years (Schmit et al. 2024). In Figure 13 it varies from 0.5 to 3 (default is 1.00).

Because R21's half-life is estimated to be about five years, it has little effect on lifetime averted deaths and efficacy Figure 13. It has negligible effect on λ_{RH5}, but higher half-lives can substantially lower λ_{R21}.

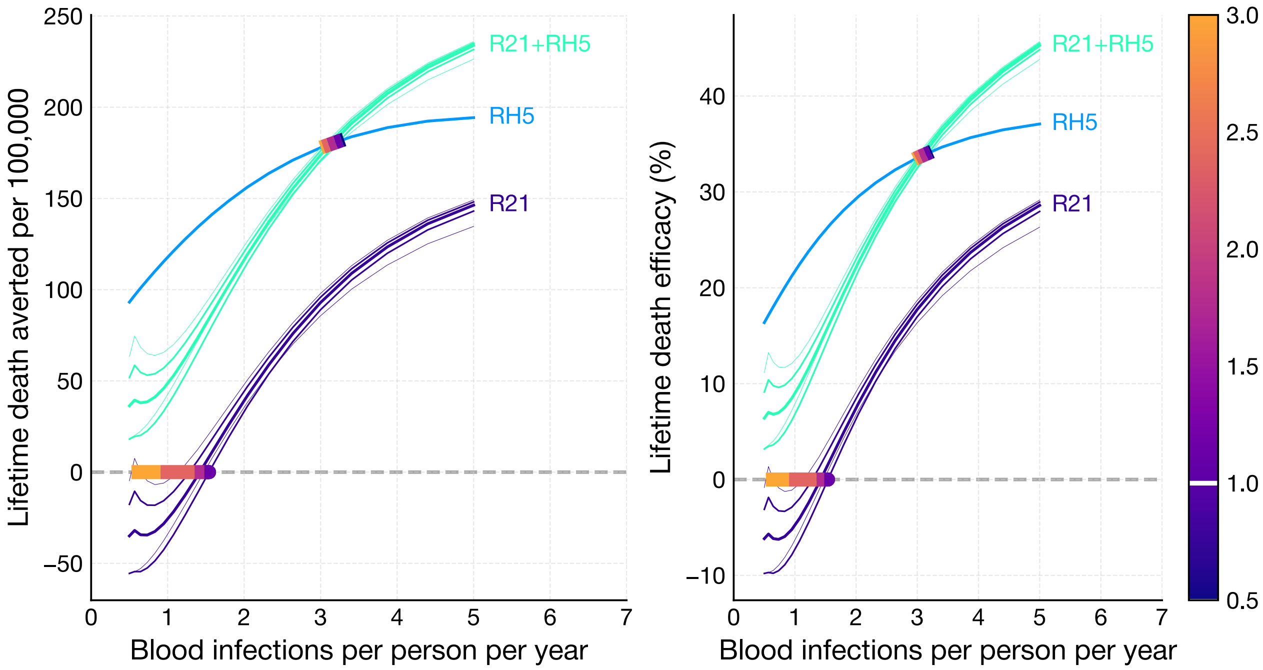

Figure 13: R21 half-life, $\tau_{\text{liver}} = (0.5, 3)$.

6.3 Number of R21 boosters

Increasing the number of R21 boosters is similar to slightly increasing its half-life: the protective efficacy of the vaccine lasts a little longer. In Figure 14 it varies from 0 to 4 (default is 1).

It has negligible effect on the critical transmission rates.

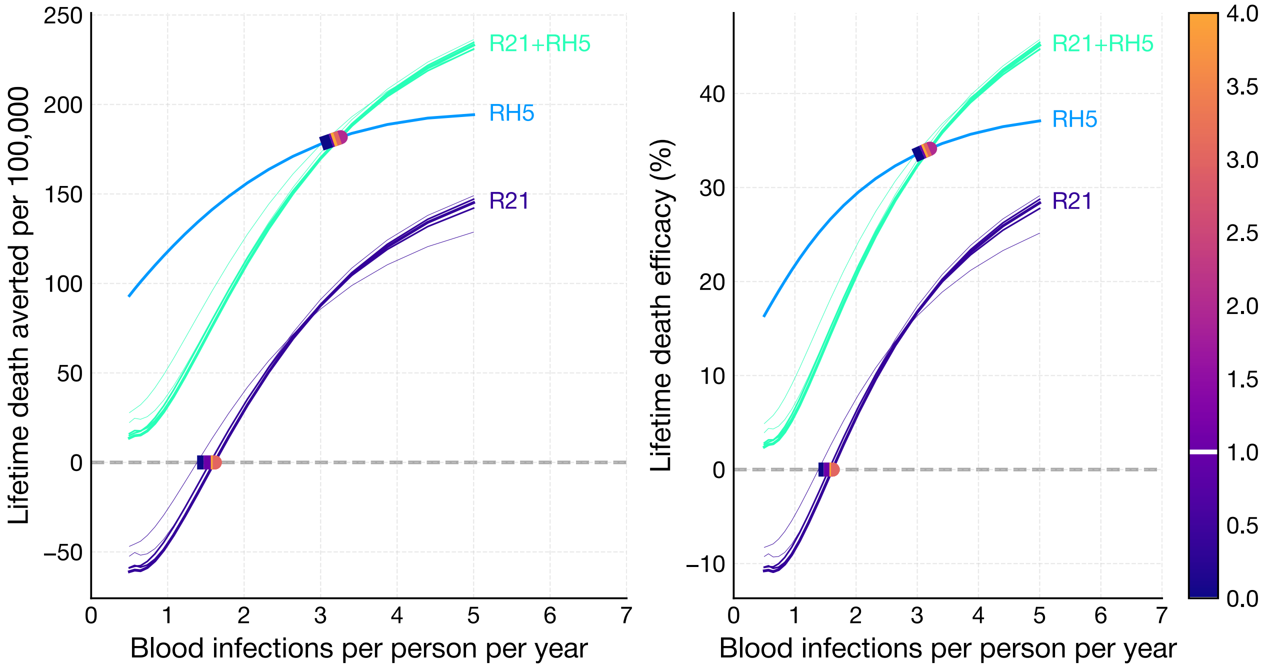

Figure 14: Number of R21 boosters = 0, …, 4.

7 RH5 vaccine parameters

The RH5 protection profile against blood infection, clinical malaria and severe malaria is assumed to mirror its anti-RH5.1 IgG antibody profile ((Silk et al. 2024), Notebook S9). There is one parameter we can change that affect this profile: the vaccine's half-life τ_{blood}. We can also change the number of yearly boosters.

Data from the phase 2b trial of the RH5.1/Matrix-M blood stage vaccine ((Natama et al. 2024)) and from primate studies demonstrate that RH5 reduces parasite growth rate and parasitaemia at clinical presentation. But the data are equivocal about the degree to which RH5 reduces the risks of blood infection, clinical malaria and severe malaria. There are three model parameters that determine these reduced risks, $\omega_{\text{infection}}$, ω_{clinical} and ω_{severe}, respectively.

7.1 RH5 half-life, τ_{blood}

The parameter τ_{blood} scales the half-life of RH5, where $\tau_{\text{blood}} = 1$ corresponds to the default parameter setting. In Figure 15 it varies from 0.5 to 3 (default is 1).

RH5 is estimated to have a short half-life of about six months. Hence increasing this parameter has a large effect on lifetime averted deaths and efficacy. It has a substantial impact on λ_{RH5}. It has no effect on λ_{R21}.

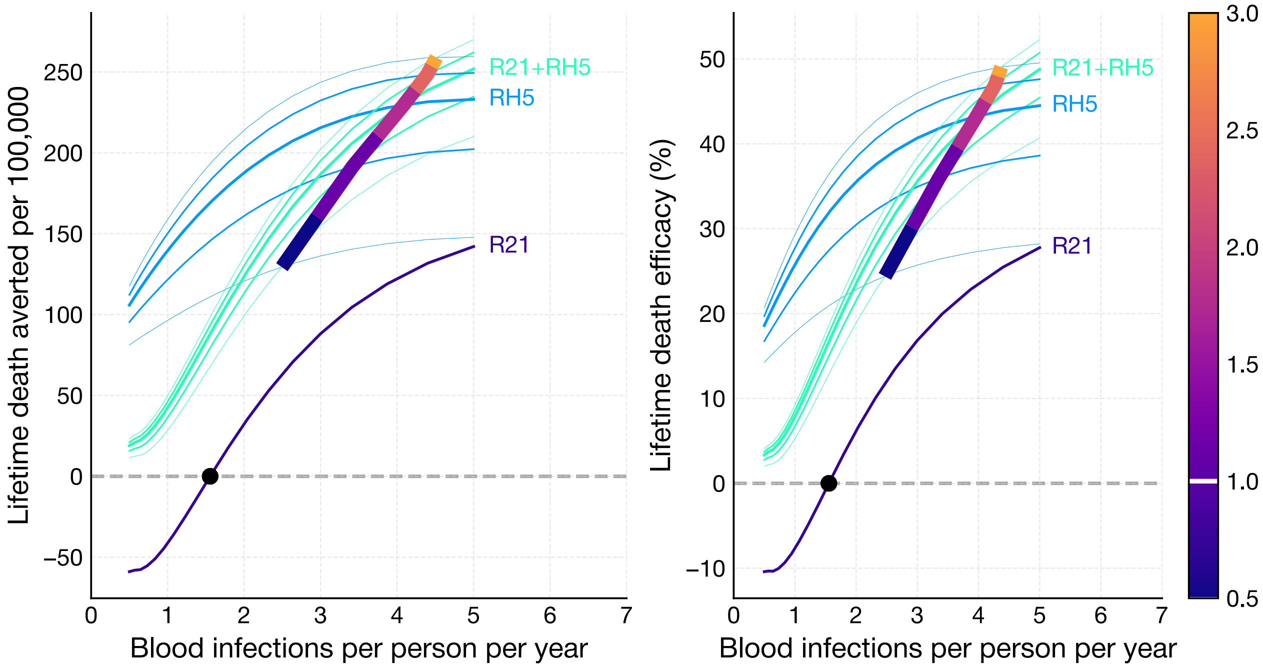

Figure 15: RH5 half-life, $\tau_{\text{blood}} = (0.5, 3)$ default 1.

7.2 Number of RH5 boosters

Increasing the number of RH5 boosters is similar to increasing its half-life: the protective efficacy of the vaccine lasts longer. In Figure 16 it varies from 0 to 4 (default is 1).

The effect on λ_{RH5} is complex, albeit small.

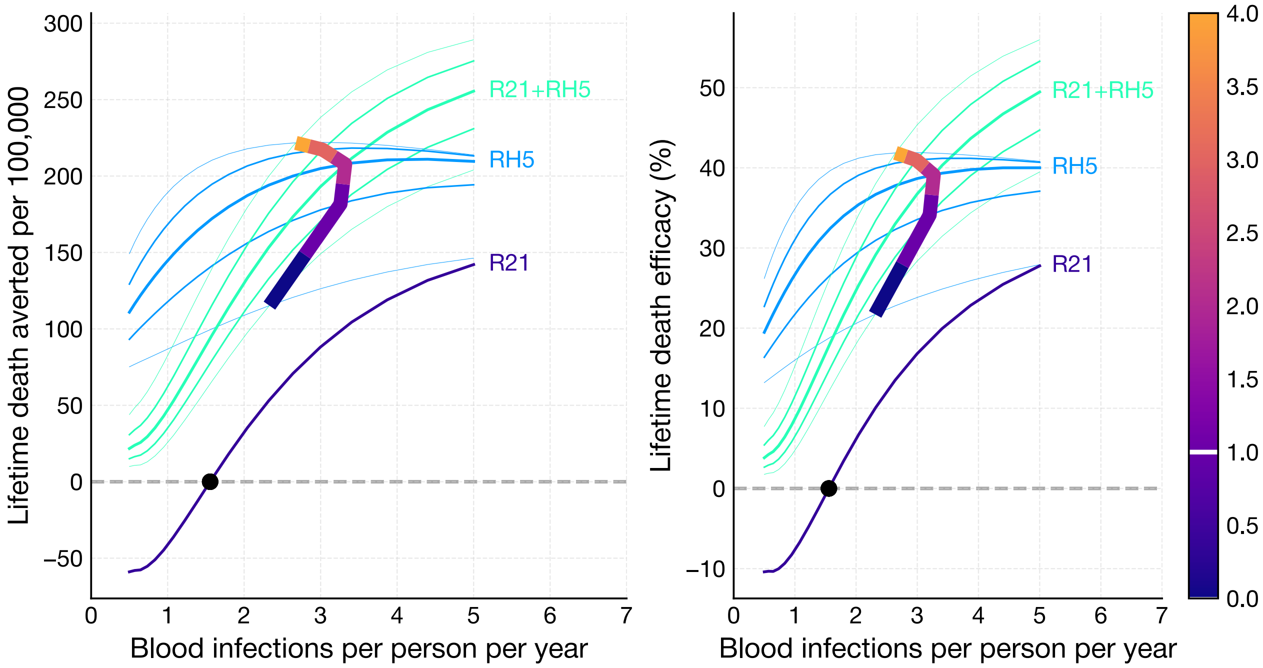

Figure 16: Number of RH5 boosters = 0, …, 4, default 1.

7.3 RH5 protective effect against blood infection, $\omega_{\text{infection}}$

RH5 protective effect against blood infection, $\omega_{\text{infection}}$, is estimated in Notebook S9. We estimated its value to be either 0% if RH5 does not protect against blood infection, to 43% if it does. In Figure 17 it varies from 0 to 1 (default 0.00).

Increasing this parameter makes RH5 behave more like R21 (with a shorter half-life). Therefore the averted deaths and efficacy curves tend towards the R21-alone curves. This has the effect of reducing averted deaths and efficacy. However, there is minimal effect on λ_{RH5}.

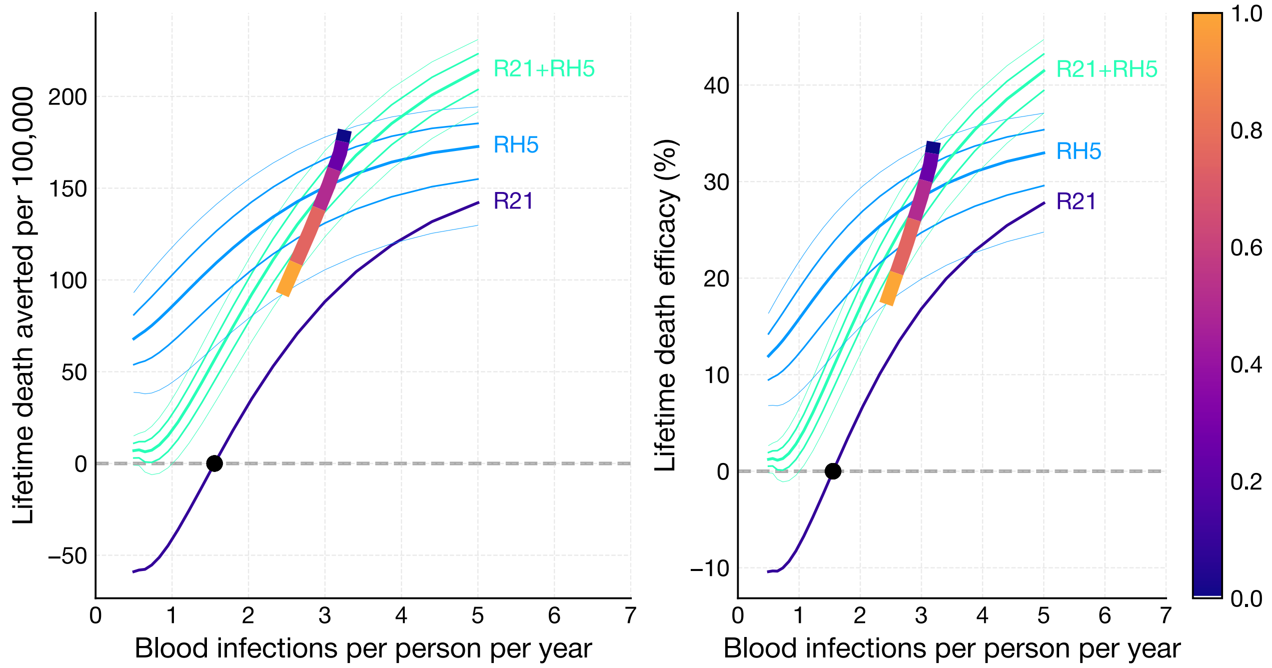

Figure 17: RH5 protective effect against blood infection, $\omega_{\mathrm{infection}} = (0, 1)$.

7.4 RH5 protective effect against clinical malaria, $\omega_{\mathrm{clinical}}$

RH5 protective effect against clinical malaria is estimated in Notebook S9. We estimated its value to be about 50% when RH5 does not protect against blood infection. In Figure 18 it varies from 0 to 0.7 (default 0.50).

The parameter reduces the risk of clinical malaria, and hence severe malaria and death. Increasing it, therefore, increases lifetime deaths averted and lifetime efficacy. It has a substantial effect on λ_{RH5}, therefore it is an important to refine its value with more precision.

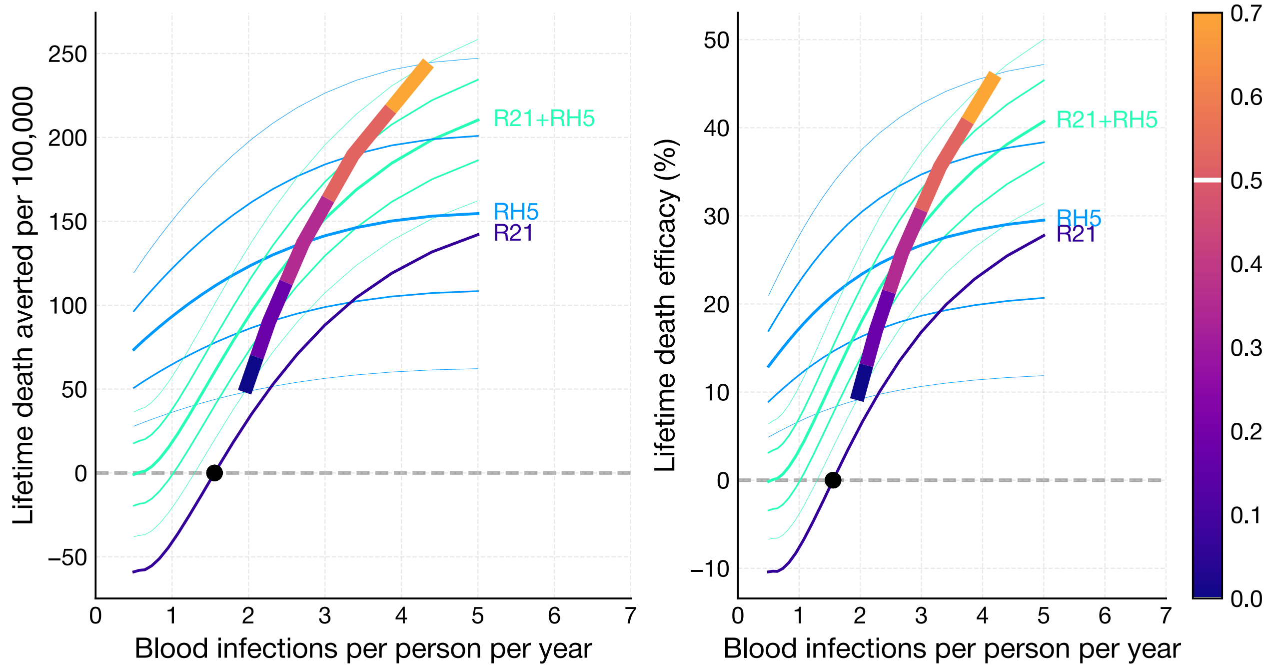

Figure 18: RH5 protective effect against clinical malaria, $\omega_{\mathrm{clinical}} = (0, 0.7)$.

7.5 RH5 protective effect against severe malaria, ω_{severe}

RH5 protective effect against severe malaria is estimated in Notebook S9. We estimated its value to be 0.29 based on fold-reduction in parasitaemia at presentation. In Figure 19 it varies from 0 to 0.5 (default 0.29).

The parameter reduces the risk of severe malaria, and hence death. Increasing it, therefore, increases lifetime deaths averted and lifetime efficacy. It has a minimal effect on λ_{RH5}.

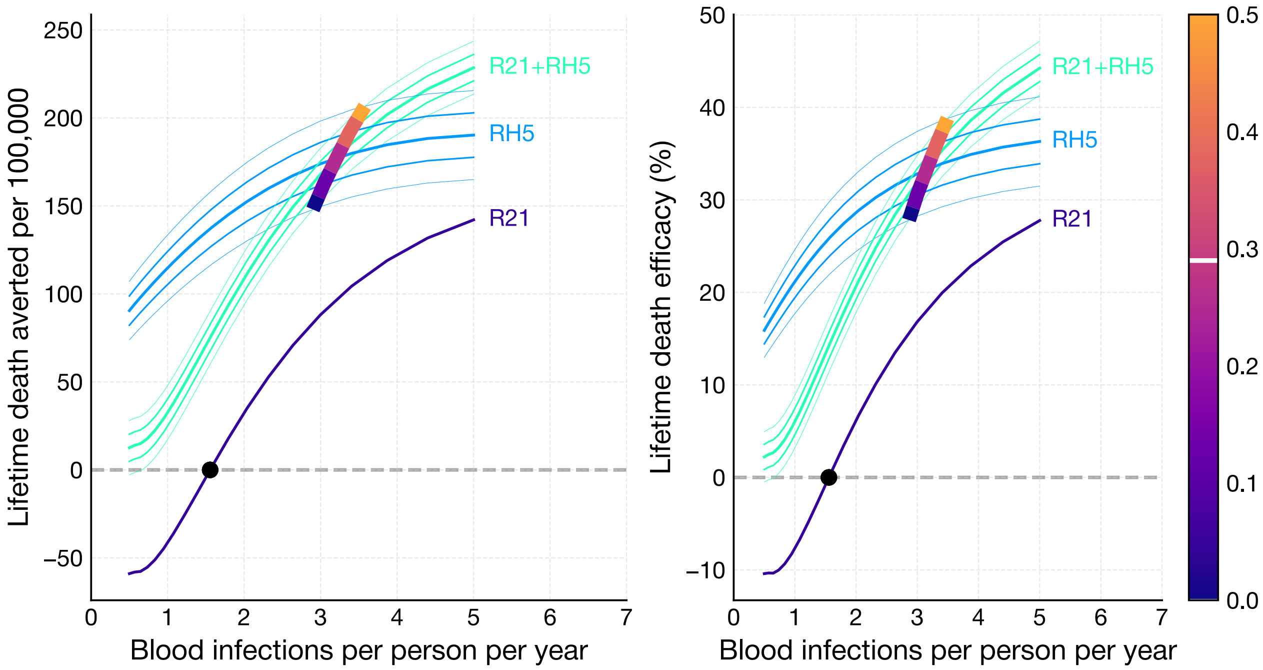

Figure 19: RH5 protective effect against severe malaria, $\omega_{\text{severe}} = (0, 0.5)$.

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